

Energy consumption estimates

(Dated: February 13, 2026)

I. INTRODUCTION

II. SIMULATION RESULTS

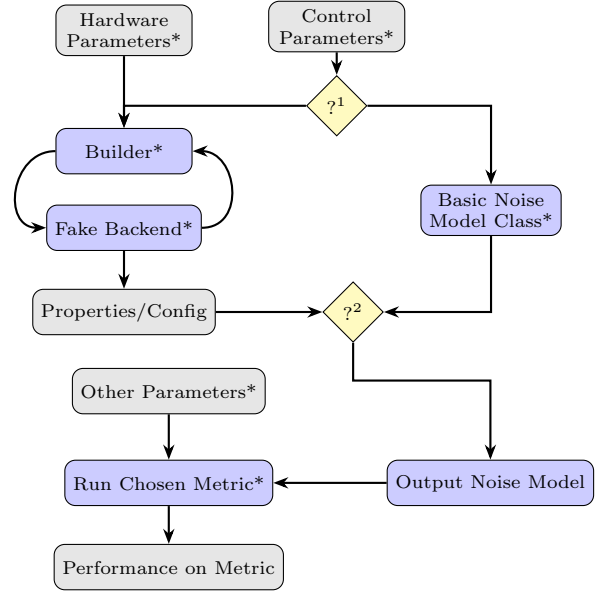
To complement this mathematical approach, we also computed simulations tracking performance on quantum algorithms as we varied temperature as the single control parameter. We computed simulations of both:

- Full quantum circuits, tracking full density matrix representations, allowing us to employ full Qiskit noise models
- Circuits using only Clifford gates, tracking qubit state using stabiliser states and only employing simpler Pauli noise models. This allowed us to simulate much larger numbers of qubits.

A. Design Architecture

Something to be emphasised is the total modularity of the system we designed to do this. All parameters can be altered and all models can be swapped in and out, providing complete flexibility over how the performance value is computed.

Here is an overview of the system architecture:



?¹: Choose between an IBM Fake Backend or a basic noise model class
 ?²: A Pauli noise model is generated if the chosen metric is a Clifford metric. If a full noise model is generated, Pauli twirling can optionally be turned off.

FIG. 1. Dataflow for simulated performance on a quantum algorithm, from control parameters to noise model to performance. All nodes noted with a * are modular and customisable.

B. Preface of Result Validity

The validity of these results hinges primarily upon the correctness of the code used to generate them. Thus, we have generated an extensive suite of unit tests to ensure the framework is operating as expected.

Moreover, the validity and applicability of these results further hinges upon the use of correct and properly catalogued assumptions used to generate them. We employ the following to mitigate the risks of compromising these qualities:

- The use of a variety of different methods of producing results, including a variety of different metrics and methods of generating noise models. These methods vary in complexity and specificity, avoiding how many assumptions are universally relied upon.
- Comprehensively cataloguing the assumptions used at every stage of generating the results, including how they change for each method of producing results.

To do this, we treat each set of results as a case study. Because there is so much overlap between the assump-

tions of each case study, we include a set of all set of assumptions in the appendix, and specify which ones apply for each.

There are a few assumptions that due to the inherent limitations of the simulation architecture universally apply, and are worth pointing out separately:

- All noise models are inherently gate-local (and thus also qubit-local)
- All noise models are state independent and unchanging over time throughout the whole metric simulation, and every shot and trial of the metric
- All noise models are Markovian

For all further assumptions, reference specific assumption sets on a per case study basis. The theory behind some of the assumptions is listed in the third subsection.

C. Case Studies

1. Case Study 1: *Quantum Volume on Legacy Backends*

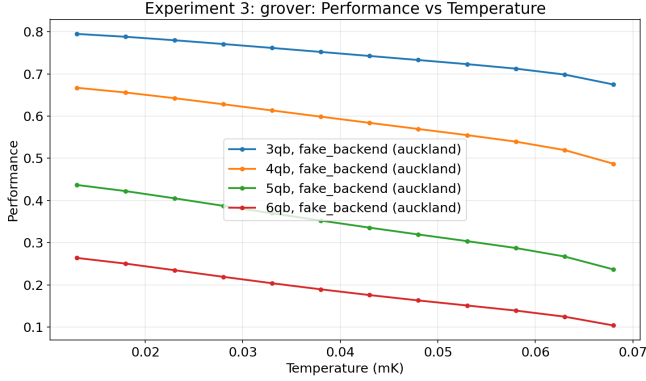


FIG. 2. Performance on Grover's algorithm using Auckland Fake Backend

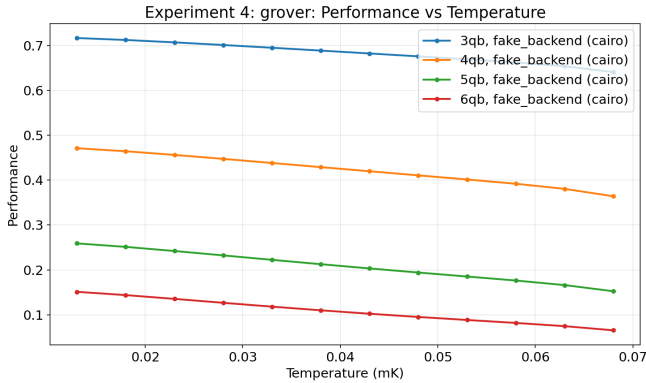


FIG. 3. Performance on Grover's algorithm using CairoV2 Fake Backend

These are both V2, Falcon R5 backends released around late 2021. You can see a stable drop in performance with increased qubit counts.

2. Case Study 1: *Quantum Volume, Simple Arrhenius Noise Model*

D. Assumptions

1. Hardware Parameters

This section outlines the modelling assumptions when creating noise models from IBM fake backends.

We implicitly use all the assumptions of the two bath model outlined in [1], Section A. In summary, we assume that the qubits only thermalise with two distinct temperature environments: the mixing chamber (MXC) and the environment.

All parameters in this model can be changed. In this paper, we choose to only vary temperature as the independent parameter of study. The rest of the parameters are either hardcoded beforehand or fit to calibration data at runtime.

To ensure the hardcoded parameters are still representative of the specific fake backend that we are modelling, for each parameter we choose one of three values, decided by the age of the backend. We characterise each backend as either modern, intermediate or legacy.

The following parameters are hardcoded beforehand:

1. $\gamma_{\text{MXC}}/\gamma_1^0 = 0.9$ (coupling ratio)
 - This is the ratio of how much of the qubit's relaxation couples to the mixing chamber (MXC) versus the residual electromagnetic environment
 - Lvov et al. [1], Table III, report experimental values of 0.60–0.85 for custom devices; we assume 0.9 as representative of IBM's professional superiority to standard lab equipment
2. The default operational cryostat temperature, assumed to be 13mK
 - All calibration data is assumed to be collected at this temperature
3. $\Delta = 2.88 \times 10^{-23} \text{ J}$ ($180 \mu\text{eV}$) (superconducting energy gap)
 - Note: This is the value for bulk aluminum value; IBM transmon electrodes and junctions use aluminum thin-films that can deviate slightly
 - With that being said, $180 \mu\text{eV}$ is standard in the superconducting qubit literature [2]
4. $\zeta = 5 \times 10^{-5}$ (subgap transparency)

- This characterizes the transparency of quasi-particle tunneling channels in the subgap regime; literature values range from 10^{-3} to 10^{-5} [3]
- It is treated as a uniform constant across the chip, and thus a manufacturing constant

The following parameters are derived on a per qubit basis at runtime.

All per-qubit data mentioned below is retrieved directly from official IBM calibration data, pulled directly from the latest github commit.

5. E_c (charging energy)

- In the transmon regime ($E_J/E_c \gg 1$), the anharmonicity approaches $\alpha \rightarrow -E_c/\hbar$ [4], Eq. (2.5)
- Since IBM reports anharmonicity $\alpha/2\pi$ in Hz:

$$E_c = h|\alpha|$$

- α represents qubit anharmonicity, and is obtained from calibration data

6. E_J (Josephson energy)

- We calculated it by inverting the transmon transition frequency $\hbar\omega_{ge} \approx \sqrt{8E_J E_c} - E_c$ [4], Eq. (2.5):

$$E_J = \frac{(\hbar\omega_{ge} + E_c)^2}{8 E_c}$$

- Qubit frequency is obtained from calibration data

7. ω_p (plasma frequency)

- Sets the energy scale for quasiparticle-induced transition rates [2], using the same equation from before from [4]:

$$\omega_p = \frac{\sqrt{8 E_J E_c}}{\hbar}$$

8. R_n (normal-state junction resistance)

- From the Ambegaokar–Baratoff relation [5], inverted using $E_J = \pi\hbar\Delta/(4e^2 R_n)$:

$$R_n = \frac{\pi \hbar \Delta}{4 e^2 E_J}$$

9. N_e (effective number of junction channels)

- Following [3], Eq. (15):

$$N_e = \frac{1}{\zeta} \cdot \frac{g_T}{2 g_K} = \frac{1}{\zeta} \cdot \frac{1}{2 R_n g_K}$$

- where $g_K = e^2/h$ is the conductance quantum
- $\zeta \sim 10^{-3}$ – 10^{-5} is the subgap transparency, a global constant set by the fabrication process
- Since R_n varies per qubit (Step 4), N_e varies per qubit

10. p_e (thermal excited-state population)

- First the probability of measuring $|1\rangle$ after preparing $|0\rangle$ is extracted for each qubit
- The residual excited state population, p_e , as estimated as equal to this value.
- This is likely a significant overestimation of p_e . Much of the readout error will not be due to excited state population, but simple measurement error.
- This in turn will likely safely overestimate T_{env} .

11. T_{env} (effective environment temperature) is estimated at runtime

- It is estimated per qubit from p_e values (Step 6) and previously derived parameters
- The median across all qubits is then taken
- Derived from the two-bath model of [1], Eq. (14), in the low- T_{MXC} limit, as described in Section II E 1

12. γ_1^0 (zero-temperature relaxation rate)

- At the operational temperature $T_{op} \approx 13$ mK, quasiparticle contributions are negligible ($\gamma_{QP}(T_{op}) \approx 0$) and the MXC bath occupation vanishes ($n_{MXC} \approx 0$)
- This permits a closed-form inversion of the two-bath relaxation model [1], Eqs. (15) and (27), as described in Section II E 2

13. $\gamma_{\varphi,base}$ (base pure dephasing rate)

- At T_{op} , quasiparticle dephasing is negligible ($\gamma_{\varphi,QP}(T_{op}) \approx 0$), so the standard decoherence decomposition [6] gives:

$$\gamma_{\varphi,base} = \frac{1}{T_{2,meas}} - \frac{1}{2 T_{1,meas}}$$

- Clamped to $\gamma_{\varphi,base} \geq 0$; a negative value would indicate $T_2 > 2T_1$, implying negligible pure dephasing
- Note: IBM reports T_2 from Hahn echo measurements. Thus, $\gamma_{\varphi,base}$ therefore captures only high-frequency dephasing sources. All T_2 predictions in this work correspond to echo coherence times

From these parameters, temperature-dependent predictions for $T_1(T)$, $T_2(T)$ are computed as described in Section II E 3.

gate fidelity $F_N(T)$

The following parameter is derived on a per gate basis at runtime:

14. ϵ_{off} (gate error offset)

- First, gate infidelity at the default operational temperature is estimated using Eq. ((12)), taking into account per qubit parameters
- The linear difference between this estimated infidelity and the actual infidelity is stored and added as an offset
- The physical interpretation of this is that this is the temperature-independent and thus temperature-stable component of gate error
- Gates without a calibrated gate error (e.g., reset gates) are assigned decoherence-only error

E. Theory

1. Environment Temperature Estimation

At the default operational cryostat temperature, the Bose–Einstein occupation of the MXC bath at qubit frequencies is vanishingly small: $n(\omega_{ge}, T_{\text{op}}) \approx 0$ at standard qubit frequencies. The two-bath effective occupation [1], Eq. (14), therefore simplifies to

$$n_{\text{eff}} \approx \frac{\gamma_{\text{env}}}{\gamma_1^0} n(\omega_{ge}, T_{\text{env}}), \quad (1)$$

Note that γ_0^1 will cancel out, so is not needed.

We can then calculate n_{eff} (effective thermal photon occupation) from the standard equation of thermal occupation of a two-level system:

$$n_{\text{eff}} = \frac{p_e}{1 - 2p_e}. \quad (2)$$

Inverting the Bose–Einstein distribution for T_{env} :

$$T_{\text{env}} = \frac{\hbar\omega_{ge}}{k_B \ln\left(1 + \frac{\gamma_{\text{env}}/\gamma_1^0}{n_{\text{eff}}}\right)}. \quad (3)$$

Combining the two equations, we get:

$$T_{\text{env}} = \frac{\hbar\omega_{ge}}{k_B \ln\left(1 + \frac{\gamma_{\text{env}}/\gamma_1^0}{p_e} (1 - 2p_e)\right)}. \quad (4)$$

Therefore, T_{env} depends only on the global coupling ratio, the per-qubit frequency, and the corrected p_e .

Using this equation, we compute T_{env} per qubit and take the median. This uses the assumption that the electromagnetic environment temperature is spatially uniform across the chip.

We use $\gamma_{\text{MXC}}/\gamma_1^0 = 0.9$.

2. Zero-Temperature Relaxation Rate

At $T_{\text{op}} \approx 13$ mK, quasiparticle-induced relaxation is exponentially suppressed ($\gamma_{\text{QP}} \propto e^{-\Delta/k_B T} \approx 0$). The total relaxation rate [1], Eqs. (15) and (27), reduces to

$$\frac{1}{T_{1,\text{meas}}} = \gamma_1^0 (2n_{\text{eff}} + 1), \quad (5)$$

where n_{eff} is computed from T_{env} (Section II E 1). Solving directly:

$$\gamma_1^0 = \frac{1}{T_{1,\text{meas}} (2n_{\text{eff}} + 1)}. \quad (6)$$

This closed-form inversion replaces numerical optimization and is exact given the low-temperature approximations.

3. Temperature-Dependent Predictions

With all per-qubit parameters from Steps 1–9, we compute the following temperature-dependent quantities at arbitrary cryostat temperature T .

a. *Quasiparticle density.* The thermal quasiparticle density follows [1]:

$$x_{\text{QP}}(T) = \sqrt{\frac{2\pi k_B T}{\Delta}} \exp\left(-\frac{\Delta}{k_B T}\right). \quad (7)$$

b. *Quasiparticle relaxation rate.* From [1], Eq. (26), originally derived in [2]:

$$\gamma_{\text{QP}}(T) = \frac{\omega_p^2}{\pi \omega_{ge}} \left[x_{\text{QP}} \sqrt{\frac{2\Delta}{\hbar\omega_{ge}}} + 4e^{-\Delta/k_B T} \cosh\left(\frac{\hbar\omega_{ge}}{2k_B T}\right) K_0\left(\frac{\hbar\omega_{ge}}{2k_B T}\right) \right] \quad (8)$$

where K_0 is the modified Bessel function of the second kind, and ω_p is per-qubit (Step 3).

c. *Quasiparticle dephasing rate.* From [1], Eq. (29), following [3]:

$$\gamma_{\varphi, \text{QP}}(T) \sim 4\pi \frac{\omega_p^2}{\omega_{ge}} \sqrt{\frac{x_{\text{QP}}^A}{N_e}}, \quad (9)$$

where $x_{\text{QP}}^A = \exp(-\Delta/k_B T)$ and both ω_p and N_e are per-qubit. We note that this expression is order-of-magnitude (as indicated by the “ $\sim$ ” in the original reference); the numerical prefactor may differ by a factor of a few.

d. *Relaxation time.*

$$T_1(T) = \frac{1}{\gamma_{\text{QP}}(T) + \gamma_1^0 (2n_{\text{eff}}(T) + 1)}, \quad (10)$$

where $n_{\text{eff}}(T) = (\gamma_{\text{MXC}}/\gamma_1^0) n(\omega_{ge}, T) + (1 - \gamma_{\text{MXC}}/\gamma_1^0) n(\omega_{ge}, T_{\text{env}})$ follows the two-bath model [1], Eq. (14).

e. Coherence time.

$$T_2(T) = \frac{1}{\frac{1}{2T_1(T)} + \gamma_{\varphi,\text{base}} + \gamma_{\varphi,\text{QP}}(T)}. \quad (11)$$

As noted in Step 9, all T_2 predictions correspond to echo coherence times.

f. Gate fidelity. For an N -qubit gate of duration t_{gate} , the decoherence-limited fidelity is [7], Eq. (6), derived from [8]:

$$F_N(T) = 1 - \frac{d t_{\text{gate}}}{2(d+1)} \sum_{i=1}^N \left(\frac{1}{T_{1,i}(T)} + \frac{1}{T_{\varphi,i}(T)} \right), \quad (12)$$

where $d = 2^N$ is the Hilbert space dimension, $T_{\varphi,i} = 1/(\gamma_{\varphi,\text{base},i} + \gamma_{\varphi,\text{QP},i}(T))$, and the sum runs over the constituent qubits of the gate. This formula assumes a local

noise model acting independently on each qubit. Equation ((12)) captures only decoherence-limited infidelity.

Temperature-independent coherent error sources are accounted for in the ϵ_{off} , a linear factor calculated using IBM calibration data.

$$\epsilon_{\text{coh}} = \epsilon_{\text{cal}} - \epsilon_{\text{dec}}(T_{\text{op}}), \quad (13)$$

where ϵ_{cal} is the calibrated gate error from the backend properties. The total predicted gate error at temperature T is then

$$\epsilon_{\text{gate}}(T) = \min(1, \epsilon_{\text{dec}}(T) + \epsilon_{\text{coh}}). \quad (14)$$

This additive decomposition of infidelities is a linear approximation; the exact result would require composing quantum channels. However, the approximation is accurate in the small-error regime ($\epsilon \sim 10^{-3}$ – 10^{-2}) where the product terms $\epsilon_{\text{dec}} \epsilon_{\text{coh}}$ are negligible. By construction, the predicted gate error recovers the calibrated value at T_{op} .

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